

AC MAINS BISTABLE SWITCH

■ T.A. BABU

This AC mains-operated bistable switch turns on or turns off a device using a miniature neon lamp and a few discrete components. This switch can be used for control panels, appliances and lighting controls.

A push-to-on switch is used to light up the neon lamp. The light emitted by the neon lamp, in turn, enables the switching action of the circuit. Use of a 555 timer wired for bistable operation makes the circuit act as a bistable switch.

The neon lamp (NL1) and the push-to-on switch (S1) are directly connected to 230V AC mains. The 12V DC supply for timer 555 (IC1) is derived from 230V AC mains through capacitive dropper C1, resistor R1 and a 12V zener diode. IC1 works as a flip-flop circuit, with the signal at its output pin 3 toggling every time it receives a pulse at its pins 2 and 6.

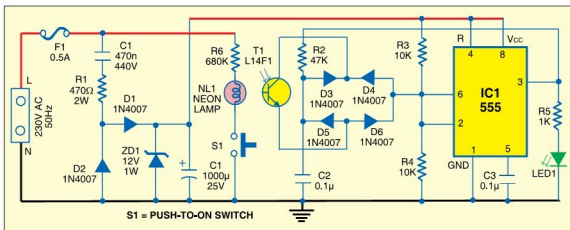
The operation of the circuit is simple. When you press switch S1 momentarily, the neon lamp glows, making phototransistor T1 conduct to provide a pulse at pins 2 and 6 of IC1. When

switch S1 is pressed, the output of IC1 goes high and LED1 glows. Pressing S1 again makes the output of IC1 low and LED1 stops glowing.

In place of LED1, you can use an opto-diode or suitable relay (not shown in the circuit) along with a suitable driver circuit to drive AC loads.

Assemble the circuit on a general-purpose PCB with the neon lamp and the phototransistor housed in a small black tube isolated from the external light source, and enclose in a suitable cabinet. Fix switch S1 on the front panel of the cabinet, and mains power cord at the rear. At the rear, also fix a 3-pin socket to connect the AC load.

Caution. Take care when operating this circuit as it is directly connected to 230V AC mains. Better still, don't attempt this circuit if you have no experience in handling high-voltage circuits. ●



LASER-BASED INTRUDER ALARM

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You can use this laser-based intruder alarm conveniently at the entrance of a hall or any other similar location to protect the same from unauthorised access as and when desired. Its range can be extended further to protect the perimeter of a building from unauthorised intrusion by using a cascade arrangement of multiple alarms.

The circuit consists of a transmitter and a receiver.

The transmitter circuit (shown in Fig. 1) is nothing but a laser diode

(LD1) driven by a 9V PP3 battery. The output of IC1 is regulated 5V as long as its input remains equal to or more than 7.5V, thus ensuring a constant drive current for the laser diode. The battery (Batt1) is connected to the circuit through switch S1.

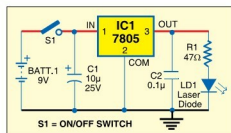


Fig. 1: Transmitter circuit

The laser diode (LD1) can be replaced with a laser pointer (torch) emitting red laser beam. The laser pointer itself can be used as a transmitter. The pointer has in-built series resistance, on/off switch and battery.

The receiver circuit (shown in Fig. 2) is basically a current-to-voltage converter built around IC LM356 (IC3). The output of IC3 is fed to the mon-